

STATIC ROUTE USING 3 ROUTERS

1. Ping 40.0.0.10 from 10.0.0.10

Static Route

TERMINAL (2)

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 5.1.2600]
(c) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  : 
   IP Address. . . . . : 40.0.0.10
   Subnet Mask . . . . . : 255.0.0.0
   Default Gateway . . . . . : 30.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>
```

2. Ping 10.0.0.10 from 40.0.0.10

Static Route

TERMINAL (2)

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 5.1.2600]
(c) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ping 10.0.0.10

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Windows IP Configuration

Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  : 
   IP Address. . . . . : 40.0.0.10
   Subnet Mask . . . . . : 255.0.0.0
   Default Gateway . . . . . : 30.0.0.1

C:\Documents and Settings\user>
```

Router and Pc configuration

R1 Configuration

Enter R1 terminal:

enable

configure terminal

hostname R1

interface ethernet0/0

ip address 10.0.0.1 255.255.255.0

no shutdown

exit

interface ethernet0/1

ip address 20.0.0.1 255.255.255.0

no shutdown

exit

ip route 40.0.0.0 255.255.255.0 20.0.0.2

end

write memory

R1 routing table

R1 knows:

10.0.0.0/24 → directly connected

20.0.0.0/24 → directly connected

40.0.0.0/24 → via 20.0.0.2

R2 Configuration

Enter R2 terminal:

enable

configure terminal

hostname R2

interface ethernet0/0

ip address 20.0.0.2 255.255.255.0

no shutdown

exit

interface ethernet0/1

ip address 30.0.0.1 255.255.255.0

no shutdown

exit

ip route 10.0.0.0 255.255.255.0 20.0.0.1

ip route 40.0.0.0 255.255.255.0 30.0.0.2

end

write memory

R2 routing table

R2 knows:

20.0.0.0/24 → directly connected

30.0.0.0/24 → directly connected

10.0.0.0/24 → via 20.0.0.1

40.0.0.0/24 → via 30.0.0.2

R3 Configuration

Enter R3 terminal:

enable

configure terminal

hostname R3

interface ethernet0/0

ip address 40.0.0.1 255.255.255.0

no shutdown

exit

interface ethernet0/1

ip address 30.0.0.2 255.255.255.0

no shutdown

exit

ip route 10.0.0.0 255.255.255.0 30.0.0.1

end

write memory

R3 routing table

R3 knows:

30.0.0.0/24 → directly connected

40.0.0.0/24 → directly connected

10.0.0.0/24 → via 30.0.0.1

Configure Win-1

On Win-1, open Command Prompt:

ipconfig

Configure:

IP Address : 10.0.0.10

Subnet Mask : 255.255.255.0

Default Gateway: 10.0.0.1

If using Windows GUI:

Settings → Network & Internet → Ethernet → IPv4 → Manual

Configure Win-2

Configure:

IP Address : 40.0.0.10

Subnet Mask : 255.255.255.0

Default Gateway: 40.0.0.1